

KIM: Knowledge-Informed Mapping (KIM) Toolkit

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Summary

We present a Knowledge-Informed Mapping toolkit in Python programming language, named KIM, to optimize the development of the mapping from a vector of inputs $\mathbf{X}$ to a vector of outputs $\mathbf{Y}$. KIM builds on the methodology development of deep learning-based inverse mapping in Jiang et al. (2023) and A. Wang et al. (2025). KIM offers a preliminary understanding of data interdependencies while optimizing the training step with uncertainty accounted for. We expect this toolkit will be helpful to glue the model data integration for Earth science applications.

Statement of need

Striving for scientific hypothesis testing and discovery, Earth scientists oftentimes develop data-driven mappings – either for inverse modeling, as part of model calibration, or forward modeling, as an emulator. Both approaches benefit from an efficient way of mapping, that projects from a vector of inputs $\mathbf{X}$ to a vector of outputs $\mathbf{Y}$. Such mapping approach has seen successes in addressing inverse and forward problems in multiple studies across Earth sciences (Cromwell et al., 2021; HU et al., 2014; Krasnopolsky & Schiller, 2003; Mudunuru et al., 2022).

Nevertheless, constructing the mapping that connects all inputs $\mathbf{X}$ to all outputs $\mathbf{Y}$ is usually challenging due to (1) limited data/simulations for training; (2) uninformative relations between some members of $\mathbf{X}$ and $\mathbf{Y}$; and (3) the structural uncertainty of the mapping. To that, Jiang et al. (2023) and A. Wang et al. (2025) leveraged the idea of integrating scientific knowledge with deep learning (Willard et al., 2022) to develop knowledge-informed mapping (KIM). The goal of this paper is to document and open source KIM for a general public usage.

Figure 1 shows the general procedures of KIM which are detailed in the next section.

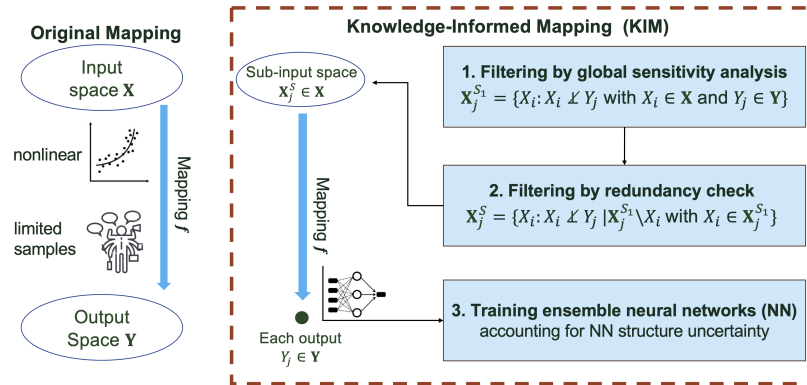


Figure 1: Comparison between KIM and the original mapping.

Mathematical approach

Consider a vector of inputs $\mathbf{X} = [X_1, \dots, X_{N_x}]$ and a vector of outputs $\mathbf{Y} = [Y_1, \dots, Y_{N_y}]$. The objective is to build up a mapping function f from $\mathbf{X}$ to $\mathbf{Y}$, such that $f: \mathbb{R}^{N_x} \rightarrow \mathbb{R}^{N_y}$, based on N_e pairs/realizations of $\mathbf{X}$ and $\mathbf{Y}$. Instead of developing a lumped mapping, we aim to develop a separate inverse mapping f_i for each $Y_j \in \mathbf{Y}$ by using a reduced space $\mathbf{X}_j^S \in \mathbf{X}$ that is most relevant to Y_j , such that $f_j: \mathbb{R}^{N_{x_j}} \rightarrow \mathbb{R}$ (see examples in Jiang et al. (2023) and A. Wang et al. (2025)), which involves the following steps.

Step 1: Filtering by global sensitivity analysis. We first perform a mutual information-based global sensitivity analysis to exclude inputs sharing zero information with Y_j and narrow down a subset $\mathbf{X}_j^{S1}$ such that:

$$\mathbf{X}_j^{S1} = \{X_i : I(X_i; Y_j) \neq 0 \text{ with } X_i \in \mathbf{X}\},$$

where $I(X_i; Y_j)$ is the mutual information between X_i and Y_j (Cover & Thomas, 2006). Based on the N_e realizations, I is calculated on the joint probability of X_i and Y_j using either binning method or k-nearest-neighbor method.

Step 2: Filtering by redundancy check. Then, we conduct a further assessment that filters out any model output in $\mathbf{X}_j^{S1}$ whose dynamics are redundant to Y_j given the knowledge of other outputs. This is achieved through a conditional independence test using conditional mutual information (Cover & Thomas, 2006) given as:

$$\mathbf{X}_j^S = \{X_i : I(X_i; Y_j | \mathbf{X}_j^{S1} \setminus X_i) \neq 0 \text{ with } X_i \in \mathbf{X}_j^{S1}\},$$

where $\mathbf{X}_j^{S1} \setminus X_i$ is the remaining set of $\mathbf{X}_j^{S1}$ by excluding X_i ; $I(X_i; Y_j | \mathbf{X}_j^{S1} \setminus X_i)$ is the conditional mutual information between X_i and Y_j conditioning on $\mathbf{X}_j^{S1} \setminus X_i$. $I(X_i; Y_j | \mathbf{X}_j^{S1} \setminus X_i) = 0$ indicates that X_i and Y_j are independent given the knowledge of $\mathbf{X}_j^{S1} \setminus X_i$.

Step 3: Uncertainty aware estimation by training ensemble neural networks. For each parameter Y_i , we train an ensemble of fully-connected neural networks by varying the hyperparameters, including the number of hidden layers, the number of hidden neurons, and the learning rate. We split the N_e model realizations into training, validation, and testing dataset. When evaluating the estimation on the test dataset, we further quantified the bias and uncertainty of the

61 prediction as:

$$\begin{aligned}\text{Bias} &= E(|\mu_w - y|) \\ \text{Uncertainty} &= E(\sigma_w/|y|),\end{aligned}$$

62 where E is the expectation operator; y is the true value; μ_w and σ_w are the mean and standard
63 deviation of the ensemble predictions weighted by their accuracy in the validation set.

64 Examples

65 We present two applications of KIM in performing inverse modeling, with Jupyter notebook
66 provided in the repository to guide the package usage. For each case, we developed three types
67 of inverse mappings: (1) the original inverse mapping without knowledge-informed, denoted as
68 M_0 ; (2) the knowledge-informed inverse mapping only using global sensitivity analysis (Step
69 1), denoted as M_1 ; and (3) the knowledge-informed inverse mapping using both Step 1 and
70 Step 2, denoted as M_2 . The configurations can be found in the example jupyter notebook.

71 **Case 1: Calibrating a cloud chamber model.** Cloud chamber model has been widely applied
72 as a virtual reality of a true cloud chamber to study turbulence, clouds, and their interactions
73 (Thomas et al., 2019; Aaron Wang, Ovchinnikov, Yang, Schmalfuss, et al., 2024; Aaron Wang,
74 Ovchinnikov, Yang, Cantrell, et al., 2024; Aaron Wang, Krueger, et al., 2024; Aaron Wang et
75 al., 2025). The objective of this example is to estimate two key parameters, i.e., the scaling
76 coefficients of wall fluxes (λ_w) and collision processes (λ_c) using inverse mapping. To that, an
77 ensemble of 513 model runs were generated based on a model set up detailed in A. Wang et al.
78 (2025), by varying the values of the two parameters using Sobol sequence. 27 virtual sensors
79 are configured, each of which 'records' multiple variables including flow properties and cloud
80 properties. The preliminary analysis and parameter estimation results are shown in Figure 2
81 and Figure 3, respectively.

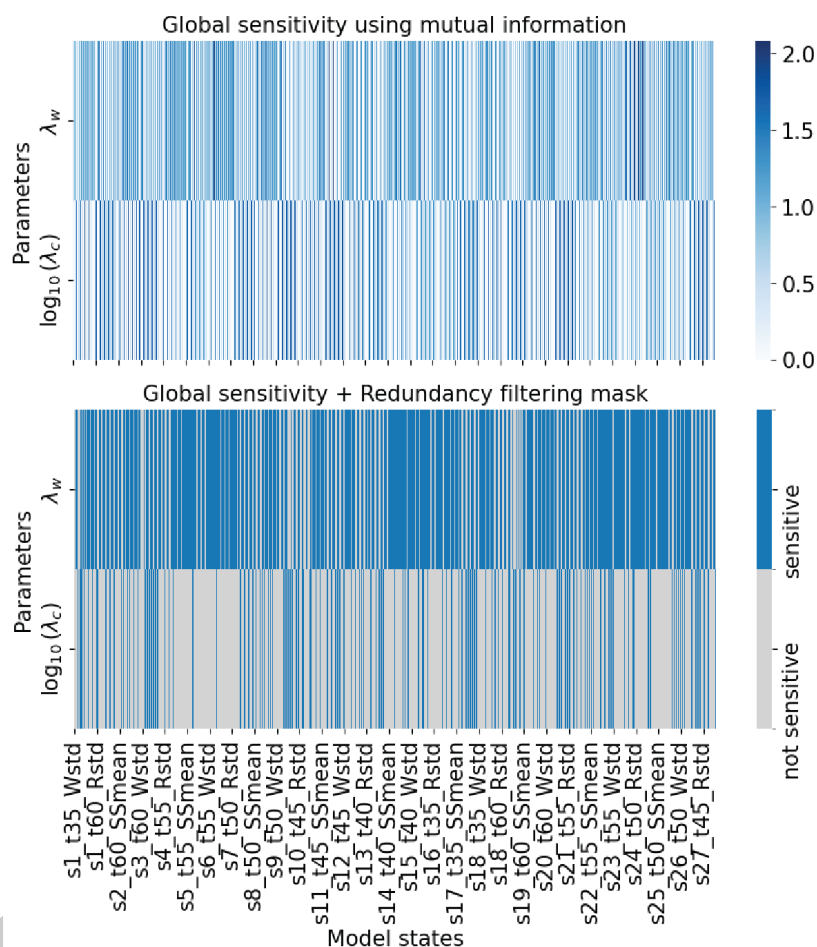


Figure 2: Preliminary analysis of cloud chamber ensemble modeling (the inputs **X**: simulated flow and cloud properties at 27 sensing locations; **Y**: two parameters of the cloud chamber model).

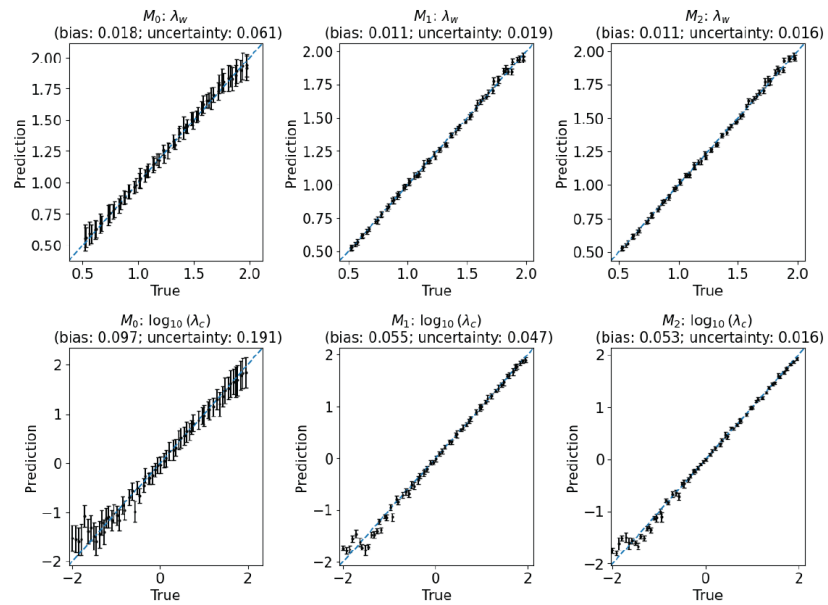


Figure 3: Estimation of the two parameters of the cloud chamber model.

Case 2: Calibrating an integrated hydrological model. The Advanced Terrestrial Simulator (ATS) is an integrated hydrological models used to simulate hydrological fluxes across a watershed (Coon et al., 2019). Here, we calibrated ATS against the streamflow observations at the outlet of Coal Creek watershed, CO, USA. The objective is to estimate eight models parameters categorized into evapotranspiration (ET), snow melting, and subsurface permeability. See Jiang et al. (2023) for more detailed information. The preliminary analysis and parameter estimation results are shown in Figure 4 and Figure 5, respectively.

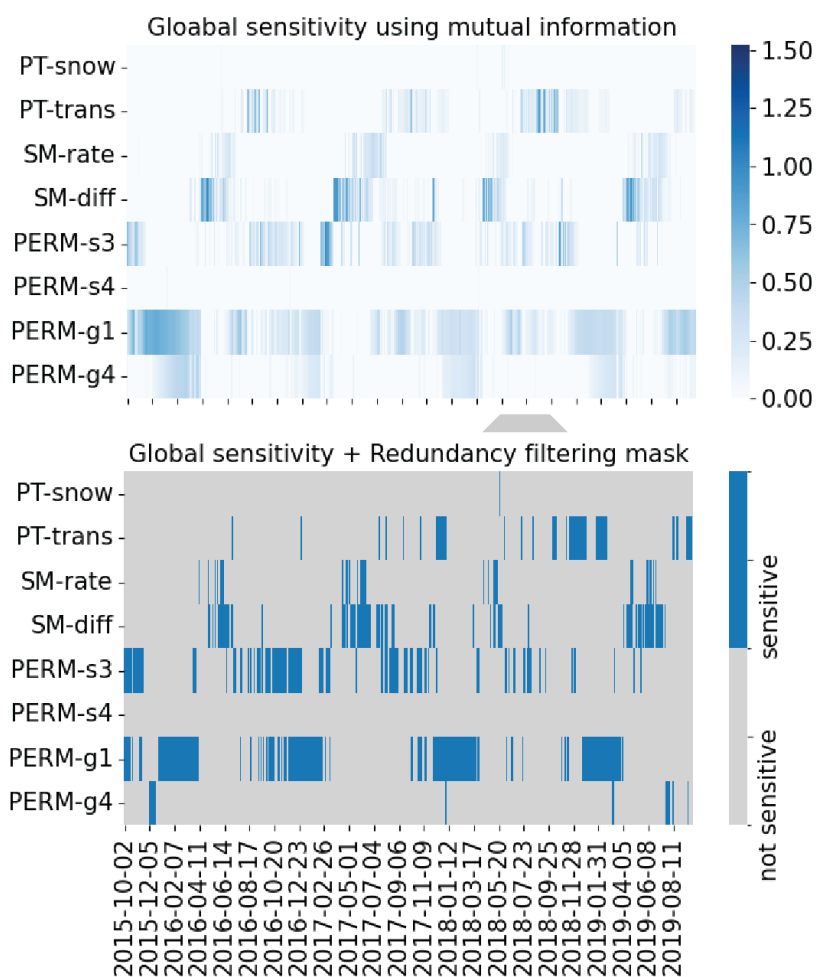


Figure 4: Preliminary analysis of ATS ensemble modeling (the inputs X : simulated streamflows; Y : eight parameters of the ATS model).

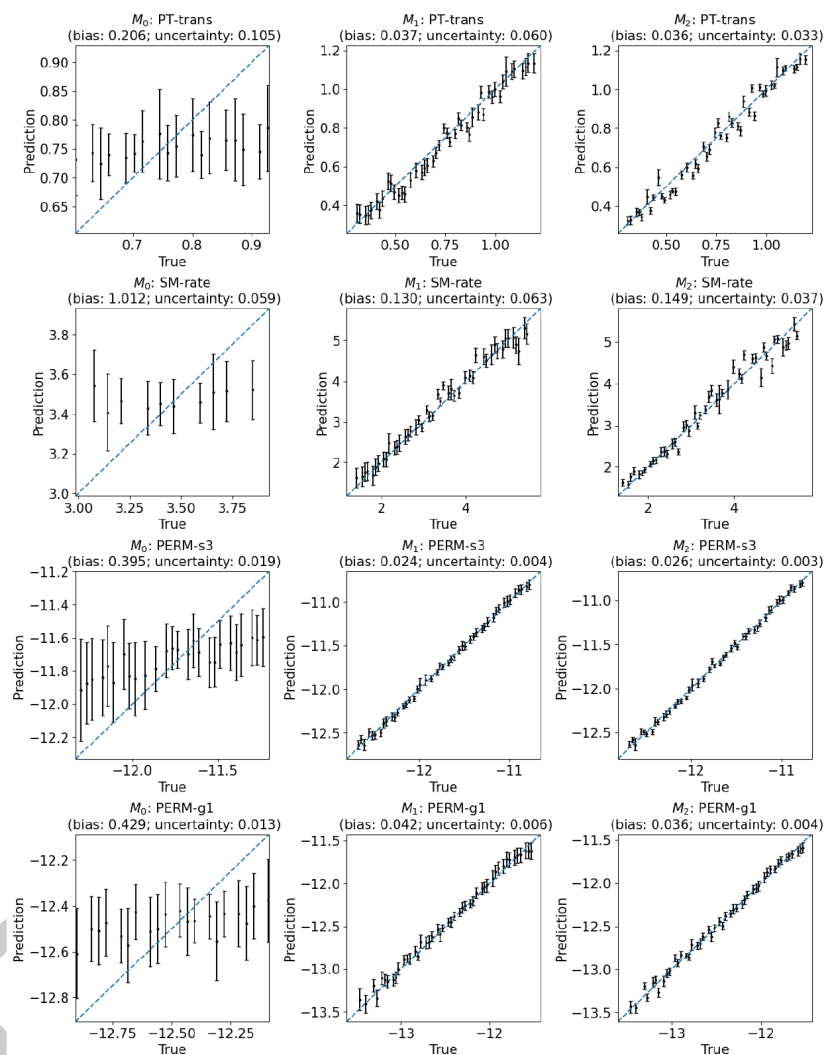


Figure 5: Estimation of four (out of eight) parameters of the ATS model.

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